Practice Set: Partial Derivatives, Gradients & Jacobians

Mathematical Foundations for Machine Learning (AIMLCZC416)

Eight multi-part problems with detailed, step-by-step solutions

Reminder of the one rule you use everywhere. To compute a partial derivative such as $\partial f/\partial x$, treat *every other variable as a constant* and differentiate as if f depended only on x. The *gradient* simply collects all the partials into one row vector, $\nabla f = [\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}]$. The *Jacobian* of a vector-valued map $\boldsymbol{f} = (f_1, \ldots, f_m)$ stacks the gradients of each component as rows, giving an $m \times n$ matrix with entry $(i, j) = \partial f_i/\partial x_j$. Everything below is just careful, repeated use of these three facts.

Problem 1. **Partial derivatives and the gradient from scratch.** Let $f(x, y) = x^2y + 3xy^2 - y^3$.

Part (a). Compute $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$, and write the gradient $\nabla f(x, y)$ as a row vector.

Part (b). Evaluate ∇f at the point $(2, -1)$.

Part (c). Verify $\dfrac{\partial f}{\partial x}$ at $(2, -1)$ directly from the limit definition $\lim_{h\to 0} \dfrac{f(2 + h, -1) - f(2, -1)}{h}$.

Solution.

Part (a). We differentiate *term by term.*

Step 1 ($\partial f/\partial x$: hold y fixed). Look at each of the three terms and differentiate with respect to x only, treating y as a number:

$$\frac{\partial}{\partial x}(x^2y) = y \cdot \frac{d}{dx}(x^2) = y \cdot 2x = 2xy, \quad \frac{\partial}{\partial x}(3xy^2) = 3y^2 \cdot \frac{d}{dx}(x) = 3y^2, \quad \frac{\partial}{\partial x}(-y^3) = 0,$$

where the last term is 0 because it contains no x. Adding the three pieces:

$$\boxed{\frac{\partial f}{\partial x} = 2xy + 3y^2}$$

Step 2 ($\partial f/\partial y$: hold x fixed). Now treat x as a number and differentiate each term with respect to y:

$$\frac{\partial}{\partial y}(x^2y) = x^2 \cdot 1 = x^2, \quad \frac{\partial}{\partial y}(3xy^2) = 3x \cdot 2y = 6xy, \quad \frac{\partial}{\partial y}(-y^3) = -3y^2.$$

Adding:

$$\boxed{\frac{\partial f}{\partial y} = x^2 + 6xy - 3y^2}$$

Step 3 (assemble the gradient). The gradient is the row vector of these two partials:

$$\nabla f(x, y) = \left[\, 2xy + 3y^2,\ \ x^2 + 6xy - 3y^2 \,\right].$$

Part (b). Substitute $x = 2,\ y = -1$ carefully (note $y^2 = 1$):

$$\frac{\partial f}{\partial x}\Big|_{(2,-1)} = 2(2)(-1)+3(1) = -4+3 = -1, \qquad \frac{\partial f}{\partial y}\Big|_{(2,-1)} = (2)^2+6(2)(-1)-3(1) = 4-12-3 = -11,$$

so $\nabla f(2,-1) = [\,-1,\ -11\,]$.

Alternate solution.

Part (c). The limit definition checks part (b) from first principles.

Step 1 (reduce to one variable). Fix $y = -1$. Then $f(x,-1) = x^2(-1) + 3x(1) - (-1) = -x^2 + 3x + 1$.

Step 2 (form $f(2+h,-1)$). Substitute $x = 2 + h$:

$$f(2+h,-1) = -(2+h)^2 + 3(2+h) + 1 = -(4+4h+h^2) + 6 + 3h + 1 = 3 - h - h^2.$$

Also $f(2,-1) = -(4) + 6 + 1 = 3$.

Step 3 (difference quotient and limit).

$$\frac{f(2+h,-1) - f(2,-1)}{h} = \frac{(3-h-h^2)-3}{h} = \frac{-h-h^2}{h} = -1 - h \xrightarrow[h\to 0]{} -1.$$

This equals the value from part (b), confirming the rule-based computation. (This limit is exactly how the lecture *defines* the partial derivative.)

Problem 2. Partial derivatives with the product rule. Let

$$f(x,y) = \frac{x^2 y}{x+2y} \qquad (x + 2y \neq 0).$$

We will *not* use the quotient rule. Instead rewrite the fraction as a product, $f(x,y) = x^2y\,(x+2y)^{-1}$, and use the product rule with the chain/power rule $\dfrac{d}{dt}\,t^{-1} = -t^{-2}$.

Part (a). Compute $\dfrac{\partial f}{\partial x}$.

Part (b). Compute $\dfrac{\partial f}{\partial y}$.

Part (c). Write ∇f and evaluate it at $(1,1)$.

Solution.

Part (a). Hold y fixed and view f as a product of two factors.

Step 1 (name the factors and their x-derivatives). Let

$$g = x^2y \;\Rightarrow\; g_x = 2xy, \qquad w = (x+2y)^{-1} \;\Rightarrow\; w_x = -(x+2y)^{-2}\cdot \underbrace{\tfrac{\partial}{\partial x}(x+2y)}_{=1} = -(x+2y)^{-2}.$$

(The chain rule produces the inner derivative $\partial(x+2y)/\partial x = 1$.)

Step 2 (apply the product rule $(gw)_x = g_x w + g w_x$).

$$\frac{\partial f}{\partial x} = 2xy\,(x+2y)^{-1} + x^2y\cdot\big(-(x+2y)^{-2}\big) = \frac{2xy}{x+2y} - \frac{x^2y}{(x+2y)^2}.$$

Step 3 (combine over the common denominator $(x+2y)^2$).

$$\frac{\partial f}{\partial x} = \frac{2xy(x+2y)-x^2y}{(x+2y)^2} = \frac{2x^2y+4xy^2-x^2y}{(x+2y)^2} = \frac{x^2y+4xy^2}{(x+2y)^2} = \frac{xy(x+4y)}{(x+2y)^2}.$$

Part (b). Now hold x fixed.

Step 1 (y-derivatives of the factors). $g = x^2y \Rightarrow g_y = x^2$, and $w = (x+2y)^{-1} \Rightarrow w_y = -(x+2y)^{-2}\cdot\underbrace{\tfrac{\partial}{\partial y}(x+2y)}_{=2} = -2(x+2y)^{-2}$.

Step 2 (product rule).

$$\frac{\partial f}{\partial y} = x^2(x+2y)^{-1} + x^2y\cdot\big(-2(x+2y)^{-2}\big) = \frac{x^2}{x+2y} - \frac{2x^2y}{(x+2y)^2}.$$

Step 3 (combine).

$$\frac{\partial f}{\partial y} = \frac{x^2(x+2y)-2x^2y}{(x+2y)^2} = \frac{x^3+2x^2y-2x^2y}{(x+2y)^2} = \frac{x^3}{(x+2y)^2}.$$

Part (c). Collect the partials and substitute $(1,1)$, where $x+2y=3$ so $(x+2y)^2=9$:

$$\nabla f = \left[\frac{xy(x+4y)}{(x+2y)^2},\ \frac{x^3}{(x+2y)^2}\right], \qquad \nabla f(1,1) = \left[\frac{1\cdot1\cdot(1+4)}{9},\ \frac{1}{9}\right] = \left[\tfrac{5}{9},\ \tfrac{1}{9}\right].$$

Alternate solution. Verify $\dfrac{\partial f}{\partial y}(1,1) = \frac{1}{9}$ from the limit definition, using only algebra.

Step 1 (reduce to one variable). Fix $x=1$: $f(1,y) = \dfrac{y}{1+2y}$, and $f(1,1) = \frac{1}{3}$.

Step 2 (difference quotient).

$$\frac{f(1,1+h)-f(1,1)}{h} = \frac{1}{h}\left(\frac{1+h}{3+2h} - \frac{1}{3}\right) = \frac{3(1+h)-(3+2h)}{3h(3+2h)} = \frac{h}{3h(3+2h)} = \frac{1}{3(3+2h)}.$$

Step 3 (limit). As $h\to0$ this $\to \dfrac{1}{3\cdot3} = \dfrac{1}{9}$, matching part (b). No differentiation rule was needed — a good way to catch product-rule algebra slips.

Problem 3. Gradient of a multivariable function with exponential and trigonometric terms. Let

$$f(x,y,z) = x^2e^{y} + y\cos z.$$

Part (a). Compute $\dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial y}, \dfrac{\partial f}{\partial z}$.

Part (b). Write ∇f and evaluate it at $(1,0,0)$.

Solution.

Part (a). There are two terms, x^2e^y and $y\cos z$. We differentiate the whole expression with respect to one variable at a time, each time treating the other two variables as constants.

Step 1 ($\partial f/\partial x$). Here e^y and $y\cos z$ are constants (no x). Only x^2 varies:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\big(x^2e^y\big) + \underbrace{\frac{\partial}{\partial x}\big(y\cos z\big)}_{=0} = e^y\cdot 2x + 0 = 2xe^y.$$

Step 2 ($\partial f/\partial y$). Now x^2 and $\cos z$ are constants. Use $\frac{d}{dy}e^y = e^y$ on the first term and $\frac{d}{dy}(y) = 1$ on the second:

$$\frac{\partial f}{\partial y} = x^2\cdot\frac{d}{dy}\big(e^y\big) + \cos z\cdot\frac{d}{dy}(y) = x^2e^y + \cos z.$$

Step 3 ($\partial f/\partial z$). Now x^2e^y is constant (no z) and y is a constant multiplier on $\cos z$:

$$\frac{\partial f}{\partial z} = \underbrace{\frac{\partial}{\partial z}\big(x^2e^y\big)}_{=0} + y\cdot\frac{d}{dz}\big(\cos z\big) = 0 + y(-\sin z) = -y\sin z.$$

Part (b). Assemble and substitute $(1,0,0)$, using $e^0 = 1,\ \cos 0 = 1,\ \sin 0 = 0$:

$$\nabla f = \big[\,2xe^y,\ x^2e^y + \cos z,\ -y\sin z\,\big],\quad \nabla f(1,0,0) = \big[\,2(1)(1),\ (1)(1)+1,\ -(0)(0)\,\big] = [2,\ 2,\ 0].$$

Alternate solution. Use linearity of the gradient: $\nabla(f+g) = \nabla f + \nabla g$, so differentiate the two terms as separate functions and add.

$$\nabla\big(x^2e^y\big) = [\,2xe^y,\ x^2e^y,\ 0\,],\qquad \nabla\big(y\cos z\big) = [\,0,\ \cos z,\ -y\sin z\,].$$

Summing the corresponding entries reproduces ∇f. Splitting a long expression into a sum of simple terms is the safest way to avoid dropping a piece.

Problem 4. Gradients involving the norm (chain rule with a radial function).

Part (a). For $f(x,y,z) = \ln\big(x^2+y^2+z^2\big)$, compute ∇f and evaluate at $(1,2,2)$.

Part (b). For $g(x,y,z) = \|\boldsymbol{x}\| = \sqrt{x^2+y^2+z^2}$, compute ∇g and evaluate at $(1,2,2)$.

Part (c). What is special about the direction and length of ∇g at any nonzero point?

Solution.

Part (a). Set $u = x^2+y^2+z^2$, so that $f = \ln u$. We use the chain rule $\dfrac{\partial}{\partial x}\ln u = \dfrac{1}{u}\cdot\dfrac{\partial u}{\partial x}$.

Step 1 (inner partials). $\dfrac{\partial u}{\partial x} = 2x,\quad \dfrac{\partial u}{\partial y} = 2y,\quad \dfrac{\partial u}{\partial z} = 2z.$

Step 2 (chain rule on each variable).

$$\frac{\partial f}{\partial x} = \frac{1}{u}\,(2x) = \frac{2x}{x^2+y^2+z^2}, \quad \text{and similarly} \quad \frac{\partial f}{\partial y} = \frac{2y}{u}, \quad \frac{\partial f}{\partial z} = \frac{2z}{u}.$$

Step 3 (assemble & evaluate). $\nabla f = \dfrac{2}{x^2+y^2+z^2}\,[\,x,\ y,\ z\,]$. At $(1,2,2)$, $u = 1+4+4 = 9$, so $\nabla f(1,2,2) = \frac{2}{9}[\,1,\ 2,\ 2\,] = \left[\frac{2}{9}, \frac{4}{9}, \frac{4}{9}\right]$.

Part (b). Now $g = u^{1/2}$, and the chain rule gives $\dfrac{\partial}{\partial x}u^{1/2} = \frac{1}{2}u^{-1/2}\dfrac{\partial u}{\partial x}$.

Step 1. $\dfrac{\partial g}{\partial x} = \frac{1}{2}u^{-1/2}(2x) = \dfrac{x}{\sqrt{u}}$, and likewise $\dfrac{\partial g}{\partial y} = \dfrac{y}{\sqrt{u}}$, $\dfrac{\partial g}{\partial z} = \dfrac{z}{\sqrt{u}}$.

Step 2 (assemble). $\nabla g = \dfrac{1}{\sqrt{u}}[\,x,\ y,\ z\,] = \dfrac{\boldsymbol{x}^\top}{\|\boldsymbol{x}\|}$.

Step 3 (evaluate). At $(1,2,2)$, $\sqrt{u} = 3$, so $\nabla g(1,2,2) = \frac{1}{3}[\,1,\ 2,\ 2\,] = \left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$.

Part (c). $\nabla g = \boldsymbol{x}/\|\boldsymbol{x}\|$ is the *unit* vector pointing radially outward from the origin. Its length is 1 everywhere: $\sqrt{(1/3)^2 + (2/3)^2 + (2/3)^2} = \sqrt{1/9 + 4/9 + 4/9} = 1$. So the distance-from-origin function increases fastest straight away from the origin, at unit rate.

Alternate solution. Treat $r = \|\boldsymbol{x}\|$ as one intermediate variable. From part (b), $\nabla r = \boldsymbol{x}^\top/r$. Since $f = \ln(r^2) = 2\ln r$, the single-variable chain rule gives $\nabla f = 2 \cdot \dfrac{1}{r}\,\nabla r = \dfrac{2}{r} \cdot \dfrac{\boldsymbol{x}^\top}{r} = \dfrac{2\boldsymbol{x}^\top}{r^2}$, the same as part (a). Reducing a radially symmetric function to a function of r alone turns a 3-variable gradient into one ordinary derivative times the known ∇r.

Problem 5. Partial derivatives by the chain (power) rule. Let $f(x,y) = \left(2x^2 - y\right)^3$.

Part (a). Compute $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

Part (b). Write ∇f and evaluate it at $(1,1)$.

Solution.

Part (a). The function has the form $(\text{inner})^3$, so we use the chain (power) rule $\dfrac{\partial}{\partial \bullet}(\text{inner})^3 = 3(\text{inner})^2 \cdot \dfrac{\partial(\text{inner})}{\partial \bullet}$, with inner $= 2x^2 - y$.

Step 1 ($\partial f/\partial x$). Hold y fixed; the inner derivative is $\partial(2x^2 - y)/\partial x = 4x$:

$$\frac{\partial f}{\partial x} = 3(2x^2 - y)^2 \cdot 4x = 12x\,(2x^2 - y)^2.$$

Step 2 ($\partial f/\partial y$). Hold x fixed; the inner derivative is $\partial(2x^2 - y)/\partial y = -1$:

$$\frac{\partial f}{\partial y} = 3(2x^2 - y)^2 \cdot (-1) = -3\,(2x^2 - y)^2.$$

Part (b). Assemble and substitute $(1,1)$, where the inner is $2(1)^2 - 1 = 1$ so $(2x^2 - y)^2 = 1$:

$$\nabla f = \left[\, 12x(2x^2 - y)^2,\ -3(2x^2 - y)^2 \,\right], \qquad \nabla f(1,1) = \left[\, 12(1)(1),\ -3(1) \,\right] = [12,\ -3].$$

Alternate solution. Factor out the common $3(2x^2 - y)^2$ to expose the structure

$$\nabla f = 3(2x^2 - y)^2 \left[\, 4x,\ -1 \,\right] = 3(2x^2 - y)^2\, \nabla(2x^2 - y).$$

That is, the gradient of $(\text{inner})^3$ is $3(\text{inner})^2$ times the gradient of the inner function — the multivariable chain rule in vector form. This factored form makes the evaluation at $(1,1)$ a one-line substitution and avoids ever expanding the cube.

Problem 6. Jacobian of a vector-valued map. Consider $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^3$,

$$\boldsymbol{f}(x,y) = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} x^2 - y^2 \\ x\, e^y \\ \ln(1 + x^2 + y^2) \end{bmatrix}.$$

Part (a). Compute the Jacobian $J_{\boldsymbol{f}}(x,y)$ and state its dimensions.

Part (b). Evaluate it at $(1,0)$.

Part (c). Say in words what the rows and columns represent.

Solution.

Part (a). The Jacobian has one row per output component and one column per input variable, so it is 3×2 with entry $(i,j) = \partial f_i / \partial x_j$. We compute the six partials individually.

Step 1 (row 1: $f_1 = x^2 - y^2$). $\dfrac{\partial f_1}{\partial x} = 2x, \qquad \dfrac{\partial f_1}{\partial y} = -2y.$

Step 2 (row 2: $f_2 = xe^y$). Treating y constant, $\dfrac{\partial f_2}{\partial x} = e^y$. Treating x constant, $\dfrac{\partial f_2}{\partial y} = x\, e^y$ (since $\frac{d}{dy} e^y = e^y$).

Step 3 (row 3: $f_3 = \ln(1 + x^2 + y^2)$). With inner function $u = 1 + x^2 + y^2$ and $\frac{\partial}{\partial x} \ln u = \frac{1}{u} \dfrac{\partial u}{\partial x}$:

$$\frac{\partial f_3}{\partial x} = \frac{2x}{1 + x^2 + y^2}, \qquad \frac{\partial f_3}{\partial y} = \frac{2y}{1 + x^2 + y^2}.$$

Step 4 (assemble).

$$J_{\boldsymbol{f}}(x,y) = \begin{bmatrix} 2x & -2y \\ e^y & x\, e^y \\ \dfrac{2x}{1 + x^2 + y^2} & \dfrac{2y}{1 + x^2 + y^2} \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

Part (b). Substitute $x = 1,\ y = 0$ (so $e^0 = 1$ and $1 + x^2 + y^2 = 2$), entry by entry:

$$J_{\boldsymbol{f}}(1,0) = \begin{bmatrix} 2(1) & -2(0) \\ e^0 & (1)e^0 \\ \frac{2(1)}{2} & \frac{2(0)}{2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Part (c). *Row i* is the gradient ∇f_i of the i-th output (a 1×2 row telling how output i responds to each input). *Column j* is the vector $\partial \boldsymbol{f}/\partial x_j$ (how the *whole* output vector responds when input x_j alone changes). Here column 1 is $\partial \boldsymbol{f}/\partial x$ and column 2 is $\partial \boldsymbol{f}/\partial y$.

Alternate solution. Build the matrix row-by-row as a stack of gradients: $\nabla f_1 = [2x, -2y]$, $\nabla f_2 = [e^y,\, xe^y]$, $\nabla f_3 = \left[\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}\right]$, then stack. You get the identical 3×2 matrix, confirming the computation.

Problem 7. Jacobian and invertibility of the exponential map. Consider $\boldsymbol{T} : \mathbb{R}^2 \to \mathbb{R}^2$,

$$\boldsymbol{T}(x, y) = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} e^x \cos y \\ e^x \sin y \end{bmatrix}.$$

Part (a). Compute the Jacobian $J_{\boldsymbol{T}}(x, y)$.

Part (b). Evaluate $\det J_{\boldsymbol{T}}$.

Part (c). Where is $\boldsymbol{T}$ *not* locally invertible? Is $\boldsymbol{T}$ globally one-to-one?

Solution.

Part (a). We need the four partials of $u = e^x \cos y$ and $v = e^x \sin y$. In each case e^x is differentiated as usual w.r.t. x, while w.r.t. y it is a constant multiplier.

Step 1 (partials of u). $\dfrac{\partial u}{\partial x} = e^x \cos y$ (since $\frac{d}{dx} e^x = e^x$); $\dfrac{\partial u}{\partial y} = e^x(-\sin y) = -e^x \sin y$.

Step 2 (partials of v). $\dfrac{\partial v}{\partial x} = e^x \sin y$; $\dfrac{\partial v}{\partial y} = e^x \cos y$.

Step 3 (assemble).

$$J_{\boldsymbol{T}} = \begin{bmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{bmatrix} = \begin{bmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix}.$$

Part (b). For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det = ad - bc$. Here

$$\det J_{\boldsymbol{T}} = \big(e^x \cos y\big)\big(e^x \cos y\big) - \big(-e^x \sin y\big)\big(e^x \sin y\big) = e^{2x} \cos^2 y + e^{2x} \sin^2 y = e^{2x}\big(\cos^2 y + \sin^2 y\big) = e^{2x}.$$

Part (c).

Step 1 (local invertibility). The inverse function theorem says $\boldsymbol{T}$ is locally invertible wherever $\det J_{\boldsymbol{T}} \neq 0$. Since $e^{2x} > 0$ for *every* real x, the determinant is never zero, so $\boldsymbol{T}$ is locally invertible **everywhere**.

Step 2 (global injectivity). "Locally invertible everywhere" does *not* imply globally one-to-one. Because cos and sin have period 2π in y, $\boldsymbol{T}(x, y) = \boldsymbol{T}(x,\, y + 2\pi)$ for all (x, y): distinct inputs map to the same output, so $\boldsymbol{T}$ is **not** globally one-to-one.

Alternate solution. View it through complex analysis. Writing $z = x + iy$, the map is $w = e^z$, with complex derivative $\frac{dw}{dz} = e^z$. For a holomorphic map the real Jacobian determinant equals $\left|\frac{dw}{dz}\right|^2 = |e^z|^2 = e^{2x}$, matching part (b). Since e^z never equals 0, the determinant is always positive — a clean contrast with maps (e.g. polar coordinates) whose Jacobian determinant *can* vanish.

Problem 8. Vector gradient identities and a non-symmetric quadratic. For $\boldsymbol{x} \in \mathbb{R}^n$, three identities are worth memorizing:

$$\nabla(\boldsymbol{b}^\top \boldsymbol{x}) = \boldsymbol{b}^\top, \qquad \nabla(\boldsymbol{x}^\top \boldsymbol{x}) = 2\boldsymbol{x}^\top, \qquad \nabla(\boldsymbol{x}^\top A\boldsymbol{x}) = \boldsymbol{x}^\top\big(A + A^\top\big).$$

Now let

$$f(\boldsymbol{x}) = \boldsymbol{x}^\top A\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}, \qquad A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \ (\textit{not} \text{ symmetric}), \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Part (a). Write f explicitly in components (x_1, x_2).

Part (b). Apply the identities to get ∇f. Why does $A + A^\top$ (not $2A$) appear?

Part (c). Evaluate ∇f at $(1, 1)$.

Solution.

Part (a). Expand the quadratic form step by step.

Step 1 (compute $A\boldsymbol{x}$). $A\boldsymbol{x} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ 3x_2 \end{bmatrix}$.

Step 2 (compute $\boldsymbol{x}^\top A\boldsymbol{x}$). $\boldsymbol{x}^\top(A\boldsymbol{x}) = [x_1, x_2] \begin{bmatrix} x_1 + 2x_2 \\ 3x_2 \end{bmatrix} = x_1(x_1 + 2x_2) + x_2(3x_2) = x_1^2 + 2x_1x_2 + 3x_2^2$.

Step 3 (subtract $\boldsymbol{b}^\top \boldsymbol{x} = x_1 + x_2$).

$$f(x_1, x_2) = x_1^2 + 2x_1x_2 + 3x_2^2 - x_1 - x_2.$$

Part (b).

Step 1 (form $A + A^\top$). $A^\top = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$, so $A + A^\top = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}$.

Step 2 (apply the identities). Using $\nabla(\boldsymbol{x}^\top A\boldsymbol{x}) = \boldsymbol{x}^\top(A + A^\top)$ and $\nabla(\boldsymbol{b}^\top \boldsymbol{x}) = \boldsymbol{b}^\top$,

$$\nabla f = \boldsymbol{x}^\top(A + A^\top) - \boldsymbol{b}^\top = [x_1, x_2] \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix} - [1, 1] = [\,2x_1 + 2x_2 - 1, \ \ 2x_1 + 6x_2 - 1\,].$$

Step 3 (why $A + A^\top$?). Write $\boldsymbol{x}^\top A\boldsymbol{x} = \sum_{i,j} A_{ij}x_ix_j$. Differentiating with respect to x_k, the variable x_k appears both as the "$i = k$" factor (contributing $\sum_j A_{kj}x_j$) and as the "$j = k$" factor (contributing $\sum_i A_{ik}x_i$). Adding these gives the k-th entry of $\boldsymbol{x}^\top(A + A^\top)$. Only when A is symmetric ($A = A^\top$) does $A + A^\top = 2A$.

Part (c). Substitute $x_1 = x_2 = 1$:

$$\nabla f(1, 1) = [\,2 + 2 - 1, \ \ 2 + 6 - 1\,] = [\,3, \ 7\,].$$

Alternate solution. Differentiate the component form from part (a) directly. From $f = x_1^2 + 2x_1x_2 + 3x_2^2 - x_1 - x_2$:

$$\frac{\partial f}{\partial x_1} = 2x_1 + 2x_2 - 1, \qquad \frac{\partial f}{\partial x_2} = 2x_1 + 6x_2 - 1,$$

identical to the identity-based result. This is the surest check that you used $A + A^\top$ and not $2A$: expanding the cross term $2x_1x_2$ by hand automatically produces the symmetric off-diagonal 2's.